

CERVICAL CYTOLOGY-HISTOLOGY CORRELATION QUALITY ASSURANCE REPORT

Per ASC Clinical Practice Committee Birdsong Guideline 2017
CAP Accreditation Checklist Requirement CYP.06600

Total Cases	Concordant	Major Discordant	HSIL PV+
130 cases analyzed	33 25.4%	41 31.5%	27.3% Target: >60%

CAP ALERT: Major discordance rate of 31.5% exceeds the CAP benchmark of less than 10%. Corrective action documentation required per CYP.06600.

PV+ ALERT: HSIL positive predictive value of 27.3% falls below the CAP target of 60%. Review HSIL overcall rate.

MAJOR FALSE NEGATIVES: 27 major undercall cases identified. All require mandatory slide review per Birdsong Section 8.

Executive Summary

A total of 130 cases were analyzed.

Section 1 - Concordance Summary

Classification of all cytology-histology pairs per ASC Birdsong concordance definitions.

Concordance_Class	Count	Percent
minor_discordant	51	39.23
major_discordant	41	31.54
concordant	33	25.38
unmapped	5	3.85

Agreement Within One Grade (Birdsong Section 5)

Percentage of pairs with exact agreement or minor disagreement only. Per Birdsong, this metric is of educational interest but not statistically meaningful due to verification bias.

Metric	Count	Percentage
Agreement within one grade (concordant + minor)	84	64.6%
Exact agreement only	33	25.4%
Major discordance (outside one grade)	41	31.5%

Section 2 - Discrepancy Bucket Summary

CAP-style discrepancy classification per Birdsong Figure 1 and Figure 3. Undercall = cytology less severe than histology. Overcall = cytology more severe than histology.

Bucket	Count	Percent
MinVar	4	3.08
MajUnd	27	20.77
MinUnd	31	23.85
Agree	33	25.38
MinOver	16	12.31
MajOver	14	10.77

Section 3 - HSIL-Focused Metrics (Birdsong Section 7)

The five statistical calculations required by Birdsong Section 7 and the ASC Clinical Practice Committee. These are the primary performance metrics for cytopathology quality assurance.

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	27.27
HSIL Pap tests with minor discrepancies	45.45
HSIL Pap tests with major false-positive discrepancies	27.27
Number of major false-negative / cytology undercall discrepancies	27.0
PV+ for HSIL Pap tests (%)	27.27

PV+ Interpretation (Birdsong Section 7a)

LOW - Below CAP target of 60%. Review HSIL overcall rate.

Extended PV+ - High-Grade Cytology Group (HSIL + ASC-H + AGC-NEO)

Per Birdsong Figure 1, HSIL, ASC-H, and AGC-NEO cytology diagnoses are grouped together for concordance assessment. The extended PV+ reflects combined performance of the entire high-grade cytology group.

Metric	Value
Total high-grade Pap tests (HSIL + ASC-H + AGC-NEO)	56
High-grade Pap tests with HSIL+ histology	35
Extended PV+	62.5%

Section 4 - Intergrade Follow-Up Rates (Birdsong Section 8)

Per Birdsong Section 8: careful review of rates of HSIL histology following cytologic interpretations of ASC-US, ASC-H, and LSIL might reveal opportunities for improvement. Elevated rates may indicate systematic cytologic undercalling.

Cytology	Total Cases	HSIL+ Follow-up	HSIL+ Rate (%)
ASC-US	13	9	69.2
ASC-H	9	7	77.8
LSIL	15	11	73.3
AGC	4	2	50.0
AGC-NEO	9	8	88.9

Clinical interpretation: ASC-US to HSIL+ rates above 15-20% or LSIL to HSIL+ rates above 20-25% may indicate undercalling patterns requiring targeted educational slide review.

Section 5 - Key Quality Signals

Summary of key performance indicators with CAP benchmark comparisons. Status reflects comparison against published CAP and ASC thresholds.

Quality Signal	Value	CAP Threshold	Status
Major discordance rate	31.5%	< 10%	EXCEEDS
Minor discordance rate	39.2%	No threshold	-
Undercall proportion	44.6%	< 15%	EXCEEDS
Overcall proportion	23.1%	No threshold	-
HSIL PV+ (positive predictive value)	27.3%	> 60%	LOW
Extended PV+ (HSIL + ASC-H + AGC-NEO)	62.5%	> 60%	OK
Agreement within one grade	64.6%	Educational interest	-
Major false negative count	27	Minimize	REVIEW

Section 6 - Visualizations

All figures generated by the CHC-QA Pipeline per ASC Birdsong Guideline 2017.

Figure 1 - Concordance Distribution

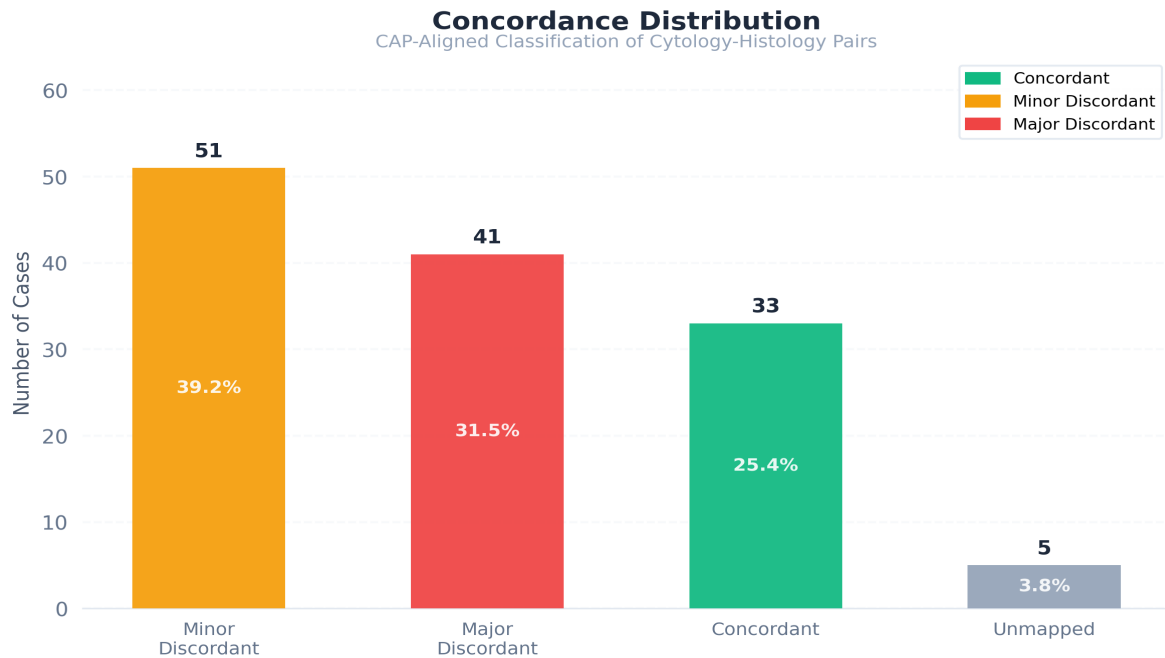


Figure 2 - Cytology-Histology Correlation Buckets

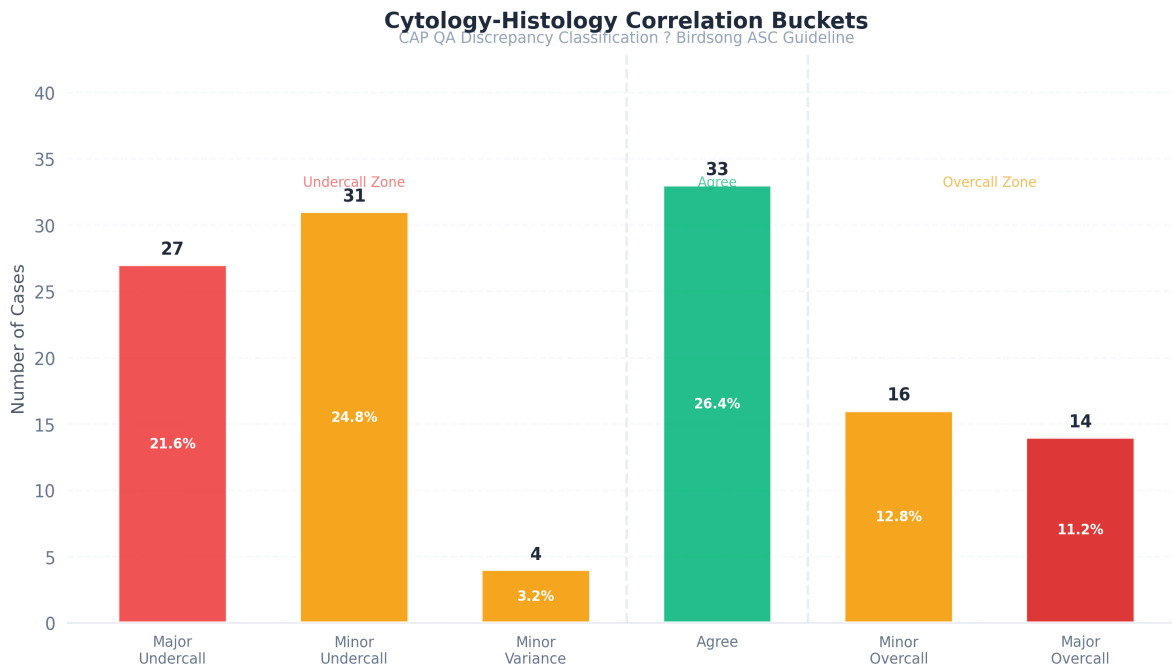


Figure 3 - HSIL Correlation Metrics

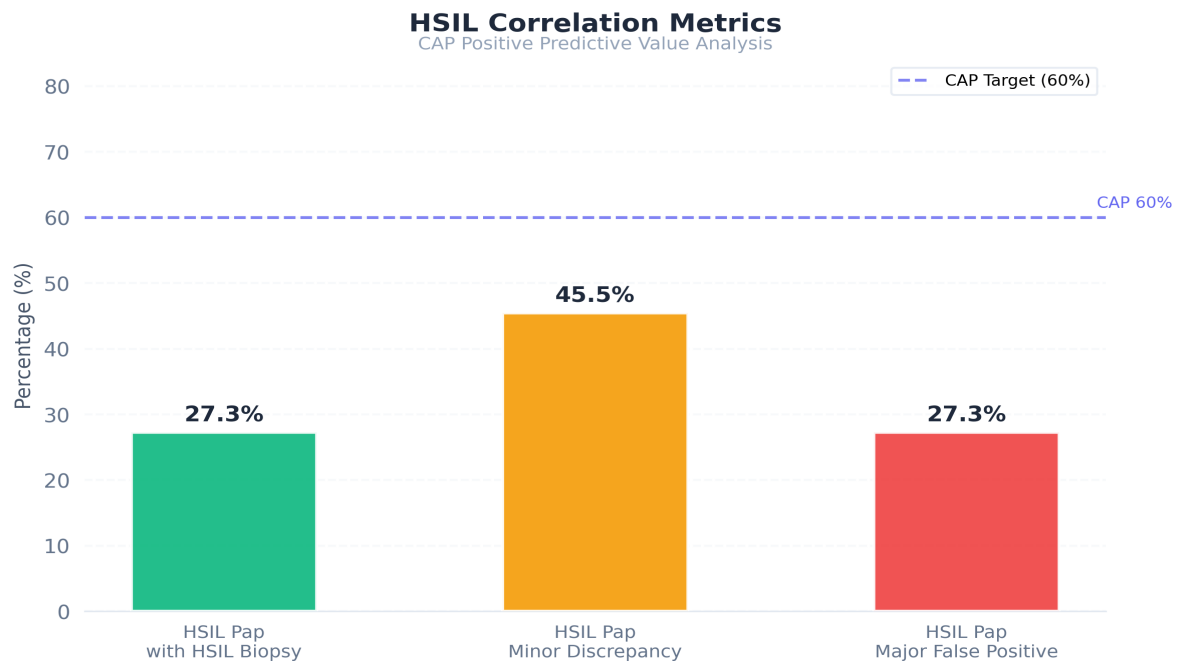


Figure 4 - Cytology vs Histology Confusion Matrix

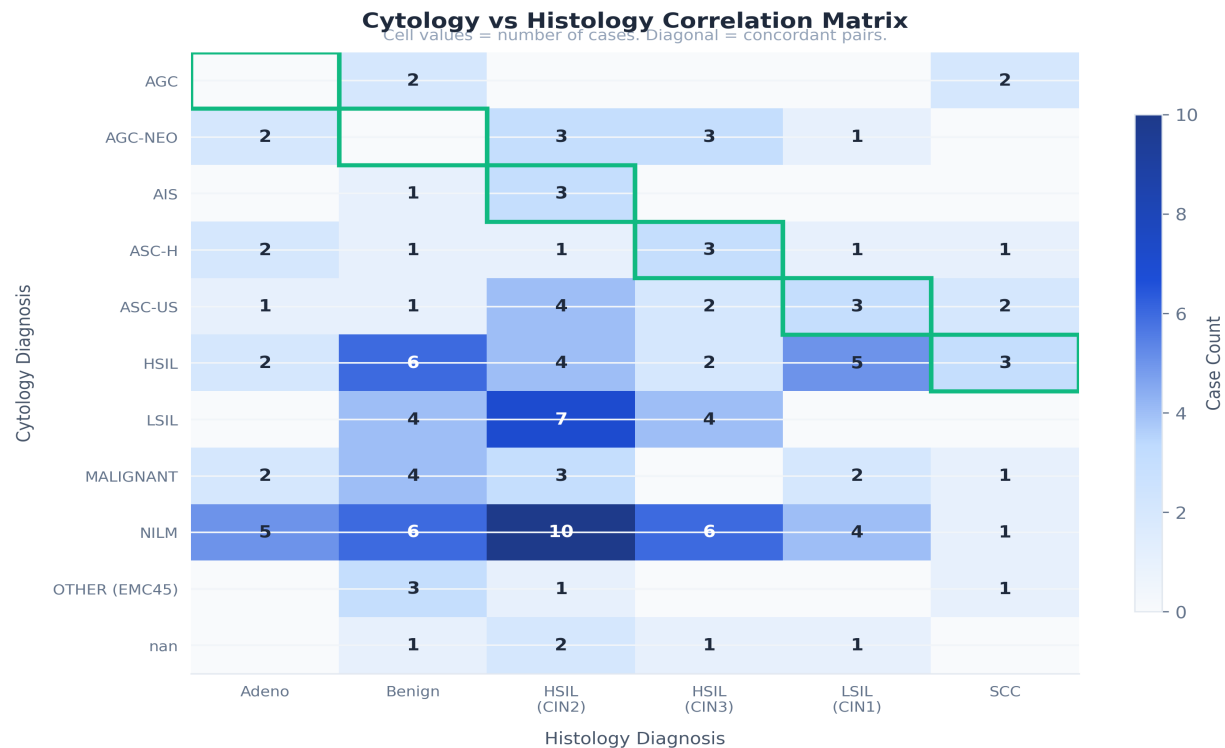
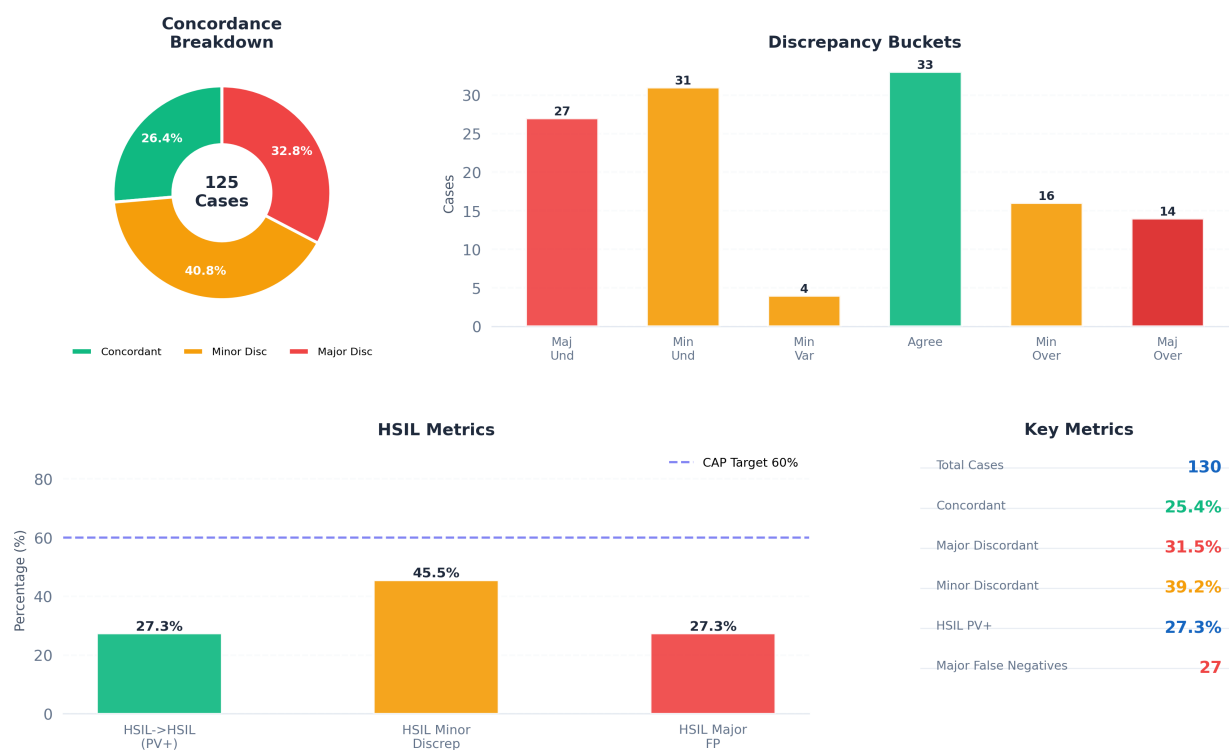


Figure 5 - Summary Dashboard

CHC-QA Pipeline | Quality Assurance Report



Section 7 - AI Clinical Commentary

The following commentary was generated by a locally-hosted AI language model based on the QA metrics above. This commentary should be reviewed by a qualified cytopathologist before use in official documentation.

Paragraph 1 - PATTERN ANALYSIS: A thorough review of the laboratory metrics reveals a significant discordance issue, with a major discordance rate of 31.5% substantially exceeding the CAP benchmark of less than 10%. The prevalence of minor discordance at 39.2% further exacerbates this problem, indicating a widespread inconsistency in diagnostic interpretations. Analysis of the data demonstrates a clear trend towards overcalling, supported by an overcall proportion of 23.1% and a notable number of major overcalls (14) and minor overcalls (16). The undercall proportion of 44.6% contributes to this imbalance, suggesting a potential systematic bias in the reporting of negative findings.

Paragraph 2 - ROOT CAUSE HYPOTHESIS: Based on Birdsong Section 8 root cause categories, interpretive error appears to be the most probable contributing factor, given the high major and minor discordance rates. The 31.5% major discordance rate suggests inconsistencies in the interpretation of cellular morphology, potentially influenced by variations in staining techniques or subjective assessment. Furthermore, the 62.5% extended PV+ rate, coupled with the 27.3% HSIL PV+ status, points to a possible issue with the identification and grading of atypical cells, particularly in complex cases. The significant number of major undercalls (27) may also reflect a tendency to underestimate the severity of lesions.

Paragraph 3 - BENCHMARK INTERPRETATION: The laboratory's performance significantly deviates from CAP benchmarks. The major discordance rate of 31.5% is substantially above the acceptable threshold, necessitating immediate corrective action. The low HSIL PV+ rate of 27.3% falls below the CAP target of 60%, indicating a potential underdiagnosis of high-grade lesions. Given these findings, a comprehensive review of laboratory protocols, including staining procedures, quality control measures, and pathologist training, is required to ensure adherence to CAP guidelines and the Australian Royal College PV+ standard of greater than 65%.

Paragraph 4 - INTERGRADE ANALYSIS: The intergrade follow-up rates highlight areas of concern, particularly regarding the HSIL+ follow-up rates across several cytology categories. The elevated HSIL+ follow-up rates for ASC-US (69.2%), LSIL (73.3%), and AGC-NEO (88.9%) suggest a potential overestimation of HSIL severity in these categories. These rates are notably higher than the expected ranges for ASC-US (15-20%) and LSIL (20-25%), raising clinical questions about the accuracy of initial diagnoses and the need for more rigorous assessment. Conversely, the lower follow-up rates for ASC-H (77.8%) and AGC (50.0%) may indicate a conservative approach to grading, potentially

Section 8 - Limitations

This report is generated by an automated deterministic cytology-histology correlation pipeline. The following limitations apply and should be considered when interpreting results.

1. **Verification bias:** CHC datasets are subject to verification bias due to selective histologic follow-up. Results should not be interpreted as measures of screening sensitivity or specificity.
2. **Clinical context:** Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.
3. **LLM normalization:** Free-text diagnosis normalization was performed by a locally-hosted large language model. Normalized terms were validated against a controlled vocabulary. Residual normalization errors may affect a small number of cases.
4. **Pairing logic:** Cases were paired by patient identifier and date window. Complex scenarios including multiple biopsies, multiple Pap tests, and delayed follow-up may result in suboptimal pairing in a minority of cases.
5. **Root cause classification:** This pipeline classifies discordance type and direction but does not perform root cause analysis (sampling error, screening error, interpretive error). Manual review of major discordant cases is required per Birdsong Section 8.

Reference: Birdsong GG, Walker JW. Gynecologic Cytology-Histology Correlation Guideline. ASC Bulletin. 2017;LIV(2):VIII-XIII. | CAP Checklist Requirement CYP.06600 | Generated by CHC-QA Pipeline (github.com/solomoneluanu/chc-qa-pipeline)